Dosage Calculation & IV Drip Rates

Formulas, worked examples and common traps for nursing medication math and NCLEX review

MindShelf

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1 Core methods

Three methods, one answer

Desired over have: $V = D/H \times Q$. D dose ordered, H dose on hand, Q volume (or tablets) holding H. **Ratio–proportion:** $H : Q = D : V$, then cross-multiply. **Dimensional analysis:** chain the order through conversion factors until only the wanted unit remains.

EXAMPLE Injection, three ways

Drug X ordered 35 mg; vial 40 mg per 1 mL.
D/H × Q: 35 ÷ 40 × 1 = 0.875 ⇒ **0.88 mL**
Ratio 40 : 1 = 35 : V and DA agree. Use a 1 mL syringe (0.01 mL marks).

TRAP

Divide only once units match: g against mg is off by a factor of 1,000.

2 Unit conversions

The ×1,000 ladder

g → mg → mcg, L → mL: × 1,000 per step down, ÷ per step up. lb → kg: × 0.45359237 (exact; courses ÷ 2.2). in → cm: × 2.54 (exact).

US measure	Exact (NIST)	Label & course
1 tsp	≈ 4.93 mL	5 mL
1 tbsp	≈ 14.79 mL	15 mL
1 fl oz	≈ 29.57 mL	30 mL
1 cup	≈ 236.6 mL	240 mL

EXAMPLE Child's weight

44 lb × 0.45359237 ≈ 19.958 ⇒ **20 kg**; course shortcut 44 ÷ 2.2 = 20 kg.

CLINICAL NOTE

Oral liquids: dose in mL with an oral syringe, never a kitchen spoon.

TRAP

Pounds typed as kg: the dose is about 2.2 times too high.

3 Weight-based & safe dose

dose = mg/kg × kg; **per day** ÷ doses per day = **per dose**. Safe if low end × kg ≤ ordered daily total ≤ high end × kg.

EXAMPLE Safe-range check (Drug Y, fictional range)

20–40 mg/kg/day; 20 kg; ordered 150 mg every 8 h. Min 20 × 20 = 400 mg/day; max 40 × 20 = 800 mg/day. Ordered 150 × 24/8 = 450 mg/day: safe. Outside the range: hold, call the prescriber.

TRAP

mg/kg/day is a 24 h total: divide by doses per day first.

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4 IV infusion rates

Pump: rate (mL/h) = *V* (mL) ÷ *T* (h). **Gravity:** drip (gtt/min) = *V* × DF ÷ (*T* × 60). DF (gtt/mL) is printed on the tubing: macrodrip 10, 15 or 20; microdrip 60. **60-drop shortcut:** with microdrip, gtt/min = mL/h. **End time** = start + *V* ÷ rate.

EXAMPLE Same order, pump vs gravity
1,000 mL over 8 h. Pump: 1,000 ÷ 8 = 125 mL/h.
Gravity, DF 20: 1,000 × 20 ÷ (8 × 60) ≈ 41.67 ⇒ **42 gtt/min**.

EXAMPLE End time and microdrip
750 mL at 100 mL/h from 09:00: 750 ÷ 100 = 7.5 h (7 h 30 min) → ends 16:30. Microdrip: 100 × 60 ÷ 60 = 100 gtt/min.

TRAP
gtt/min needs minutes: hours alone give a rate 60 times too high.

5 Body surface area (Mosteller)

$$\text{BSA (m}^2\text{)} = \sqrt{\text{height (cm)} \times \text{weight (kg)} \div 3{,}600}$$

EXAMPLE BSA-based dose (Drug Z)
150 cm, 54 kg: $\sqrt{150 \times 54 \div 3{,}600} = \sqrt{2.25}$ = 1.5 m². Ordered 100 mg/m²: 1.5 × 100 = 150 mg.

TRAP
3,600 assumes cm and kg: convert inches and pounds first.

6 Reconstitution & strength

concentration = drug ÷ **final** volume on the label. % (w/v) = g per 100 mL: 0.9% = 0.9 × 1,000 ÷ 100 = 9 mg/mL. Ratio 1:1,000 = 1 g in 1,000 mL = 1 mg/mL.

EXAMPLE Powder vial (Drug X, fictional label)
1 g; label: add 4.6 mL to obtain 5 mL. 1 × 1,000 ÷ 5 = 200 mg/mL. Order 300 mg: 300 ÷ 200 = 1.5 mL (3 mL syringe, 0.1 mL marks).

TRAP
Powder adds volume: use the final 5 mL, not the 4.6 mL of diluent.

7 Titration & drips

C (mcg/mL) = mg in bag × 1,000 ÷ mL in bag. mL/h = mcg/kg/min × kg × 60 ÷ *C*. Reverse: mcg/kg/min = mL/h × *C* ÷ (kg × 60). Units: mL/h = (units/h) ÷ (units/mL).

EXAMPLE mcg/kg/min → mL/h (Drug X)
400 mg in 250 mL: 400 × 1,000 ÷ 250 = 1,600 mcg/mL. 5 mcg/kg/min, 80 kg: 5 × 80 × 60 ÷ 1,600 = 15 mL/h.

EXAMPLE units/kg/h → mL/h (Drug W)
25,000 units in 250 mL = 100 units/mL. 18 units/kg/h × 72 kg = 1,296 units/h; 1,296 ÷ 100 = 12.96 ⇒ **13 mL/h**.

CLINICAL NOTE
Heparin and insulin drips are high-alert: follow your facility's double-check policy.

TRAP
Bag in mg, dose in mcg: × 1,000 first; per-minute doses need × 60.

8 Rounding & syringes

Round once, at the end

- Keep full precision in between; weight to 0.1 kg.
- gtt/min: whole drops. mL/h: pump precision (0.1 mL/h here; some pumps: whole mL/h).

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- Syringe: to its marks, 0.01 mL (1 mL size) or 0.1 mL (3 mL size).

TRAP
Rounding the weight or the concentration early shifts the final rate.

9 Safe notation

Never write	Risk	Write
1.0 mg	Read as 10 mg (trailing zero)	1 mg
.5 mg	Read as 5 mg (no leading zero)	0.5 mg
U or u	Read as 0 or 4	unit(s)
IU	Read as IV or 10	international unit(s)
QD / QOD	Daily and every other day confused	daily / every other day
µg	Read as mg	mcg
cc	Read as u or 00	mL
MS, MSO4, MgSO4	Morphine and magnesium confused	full drug name
100000 units	Zeros miscounted	100,000 units

TRAP
A trailing zero or a missing leading zero makes a tenfold dose error.

10 Medication rights & orders

Rights (patient, drug, dose, route, time) need system safeguards, not just a checklist. **Complete order:** patient name and ID, date/time, drug, dose with unit, route, frequency, weight if weight-based, signature.

CLINICAL NOTE
Unclear or incomplete order: clarify with the prescriber before giving.

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